

PVsyst - Simulation report

Grid-Connected System

Project: Bank Grid-Tied PV System

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 23.40 kWp

Aleppo - Bank Grid-Tied PV System Site - Syria



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PVsyst V7.2.21

VC0, Simulation date:
10/07/26 10:05
with v7.2.21

Project summary

Geographical Site

Aleppo - Bank Grid-Tied PV System Site
Syria

Situation

Latitude 36.20 °N
Longitude 37.14 °E
Altitude 379 m
Time zone UTC+3

Project settings

Albedo 0.20

Meteo data

Aleppo - Bank Grid-Tied PV System Site
Meteonorm 8.0 (2006-2013), Sat=100% - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Fixed plane
Tilt/Azimuth 31 / 0 °

Near Shadings

No Shadings

User's needs

Unlimited load (grid)

System information

PV Array

Nb. of modules 39 units
Pnom total 23.40 kWp

Inverters

Nb. of units 1 unit
Pnom total 25.00 kWac
Pnom ratio 0.936

Results summary

Produced Energy 44.02 MWh/year Specific production 1881 kWh/kWp/year Perf. Ratio PR 89.30 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Orientation

Fixed plane

Tilt/Azimuth 31 / 0 °

Sheds configuration

No 3D scene defined

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

No Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer

Longi Solar

Model

LR5-72 HTH 600 M

(Custom parameters definition)

Unit Nom. Power

600 Wp

Number of PV modules

39 units

Nominal (STC)

23.40 kWp

Modules

3 Strings x 13 In series

At operating cond. (50°C)

Pmpp

21.78 kWp

U mpp

526 V

I mpp

41 A

Total PV power

Nominal (STC)

23 kWp

Total

39 modules

Module area

98.1 m²

Cell area

93.9 m²

Inverter

Manufacturer

Huawei Technologies

Model

SUN2000-25KTL-M5

(Custom parameters definition)

Unit Nom. Power

25.0 kWac

Number of inverters

2 * MPPT 50% 1 unit

Total power

25.0 kWac

Operating voltage

200-1000 V

Max. power (=>55°C)

27.5 kWac

Pnom ratio (DC:AC)

0.94

Total inverter power

Total power

25 kWac

Number of inverters

1 unit

Pnom ratio

0.94

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const) 29.0 W/m²K

Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res.

138 mΩ

Loss Fraction

1.0 % at STC

Module Quality Loss

Loss Fraction

0.0 %

Module mismatch losses

Loss Fraction 2.0 % at MPP

Strings Mismatch loss

Loss Fraction

0.1 %

IAM loss factor

Incidence effect (IAM): User defined profile

0°	25°	45°	60°	65°	70°	75°	80°	90°
1.000	1.000	0.995	0.962	0.936	0.903	0.851	0.754	0.000



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Main results

System Production

Produced Energy

44.02 MWh/year

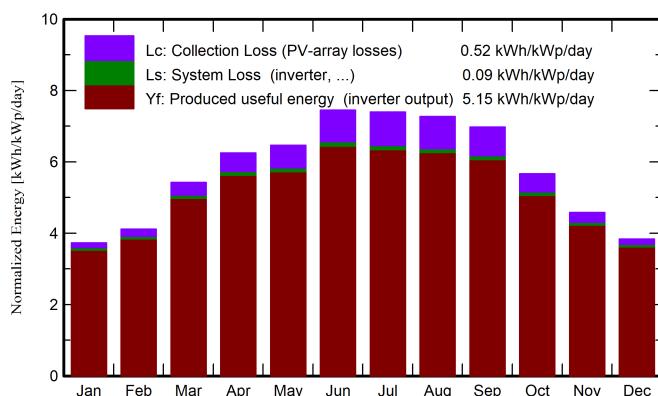
Specific production

1881 kWh/kWp/year

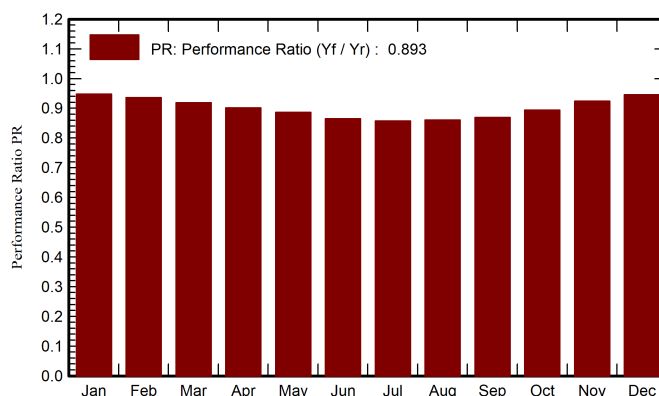
Performance Ratio PR

89.30 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	MWh	MWh	ratio
January	76.2	36.13	5.47	115.5	113.9	2.611	2.563	0.948
February	86.1	40.64	7.27	115.1	113.4	2.568	2.521	0.936
March	140.7	61.71	11.73	168.1	165.5	3.680	3.615	0.919
April	174.3	68.39	16.36	187.4	184.2	4.026	3.954	0.902
May	208.2	81.87	22.15	200.5	196.4	4.232	4.158	0.886
June	243.4	57.51	27.34	223.5	218.7	4.611	4.526	0.865
July	244.1	60.28	30.62	229.4	224.7	4.687	4.604	0.858
August	218.7	58.04	30.24	225.5	221.5	4.623	4.542	0.861
September	177.8	43.99	25.82	209.3	206.1	4.334	4.258	0.869
October	131.0	44.53	20.48	175.7	173.2	3.743	3.677	0.894
November	89.3	32.13	11.99	137.5	135.7	3.027	2.972	0.924
December	72.1	30.01	6.78	118.8	117.2	2.676	2.627	0.945
Year	1862.0	615.22	18.08	2106.4	2070.5	44.818	44.017	0.893

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

PR Performance Ratio



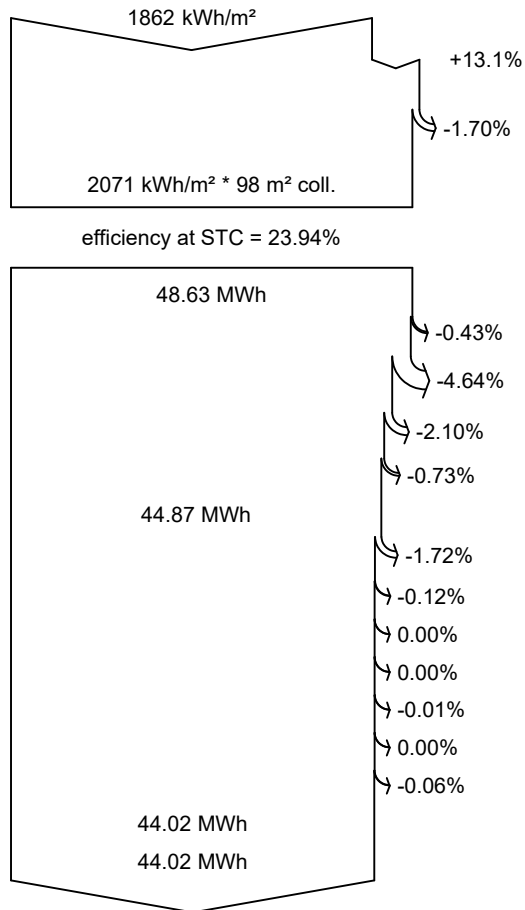
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Loss diagram



Global horizontal irradiation

Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Mismatch loss, modules and strings

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid

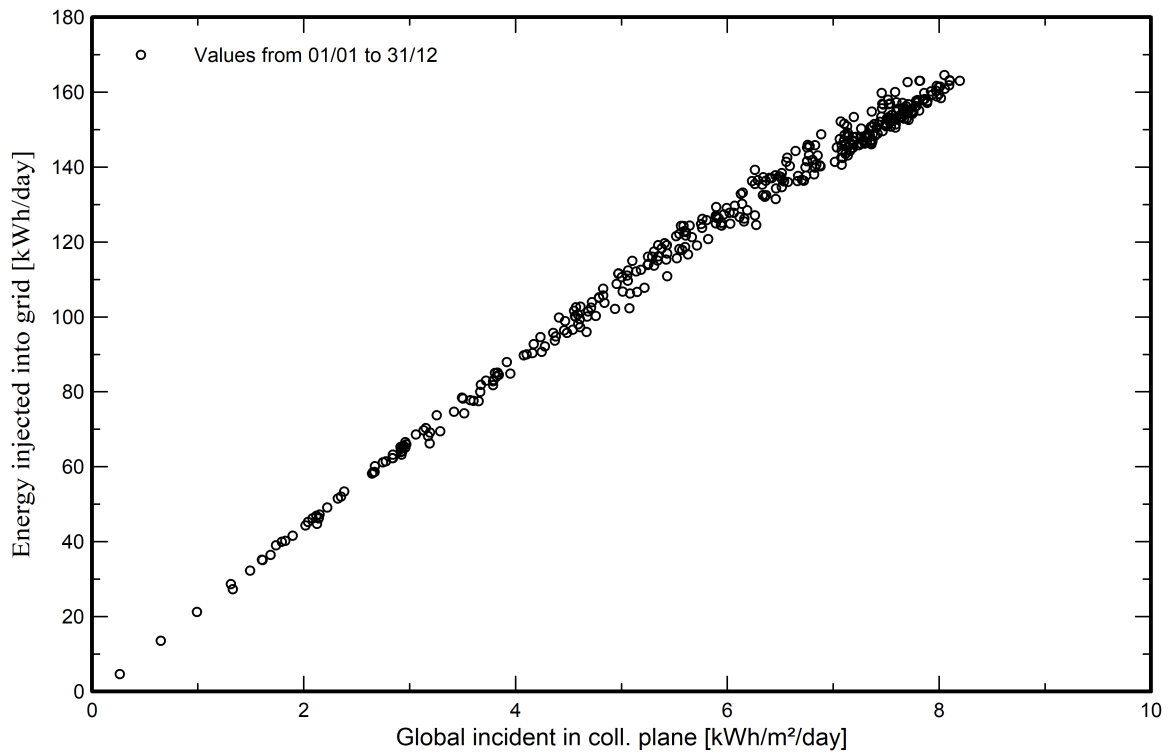


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Special graphs

Daily Input/Output diagram



System Output Power Distribution

